

SPACE PHYSICS

6.1 Earth and the Solar System

6.1.1 The Earth

1 Know that:

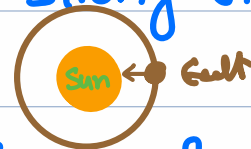
- (a) the Earth is a planet that orbits the Sun once in approximately 365 days
- (b) the orbit of the Earth around the Sun is an ellipse which is approximately circular
- (c) the Earth rotates on its axis, which is tilted, once in approximately 24 hours
- (d) it takes approximately one month for the Moon to orbit the Earth
- (e) it takes approximately 500s for light from the Sun to reach the Earth

2 Define average orbital speed from the equation

$$v = \frac{2\pi r}{T}$$

where r is the average radius of the orbit and T is the orbital period; recall and use this equation

The Earth:- Earth is the third planet from the sun and move in an elliptical orbit (oval) (nearly circular) controlled by strong Gravitational attraction of the sun.



Motion of Earth:- The Earth revolves around the Sun and also spins in its own axis. This motion of Earth is responsible for the change in seasons and for day and night.

(a) Motion around Sun:-

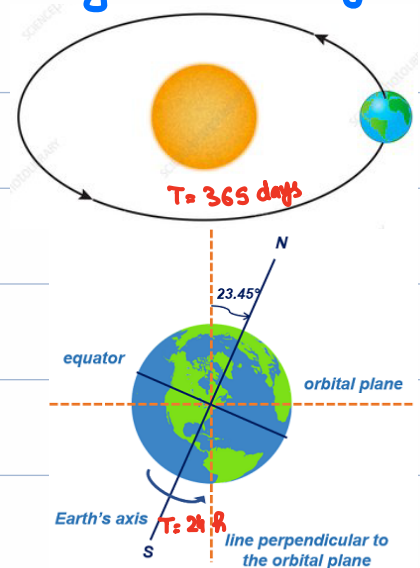
Time period of rotation: 365.25 days $\approx$ 365 days

Path: Form of an Ellipse (oval)

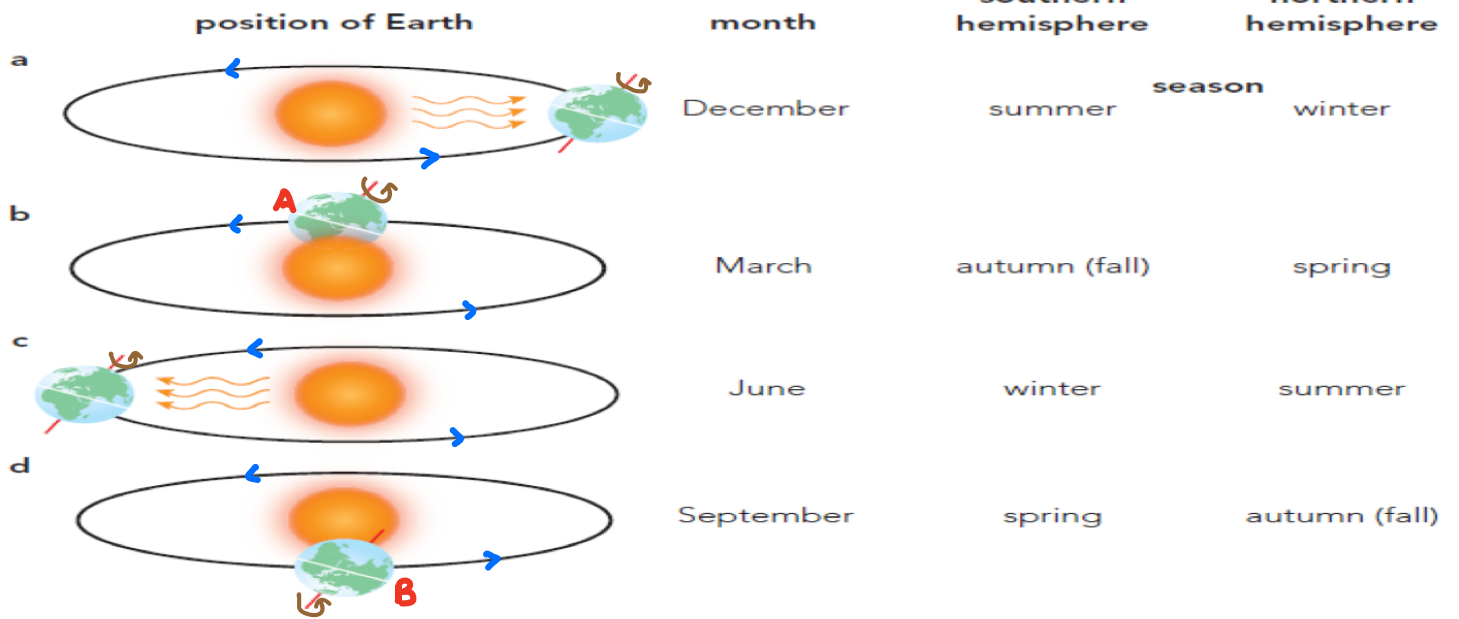
Effect: change of seasons

is due to tilt of Earth.

The Earth is tilted at an angle of approximately 23.4° with the vertical.



seasons; The Earth orbits the Sun every 365.25 days. The tilt of the Earth causes



Day and Night are of equal length only when Earth is at position A and B of its orbit – equinoxes.

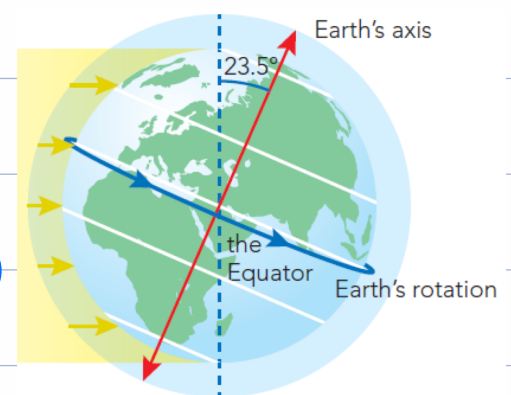
(b) Rotation of Earth:

Time period of rotation: 24 hours

Rotation: Anticlockwise at 23.4° with its line passing its poles.

Effect: Day and Night.

Day is the time when half of the Earth's surface is facing the sun and at night time, other half part is away from sun.



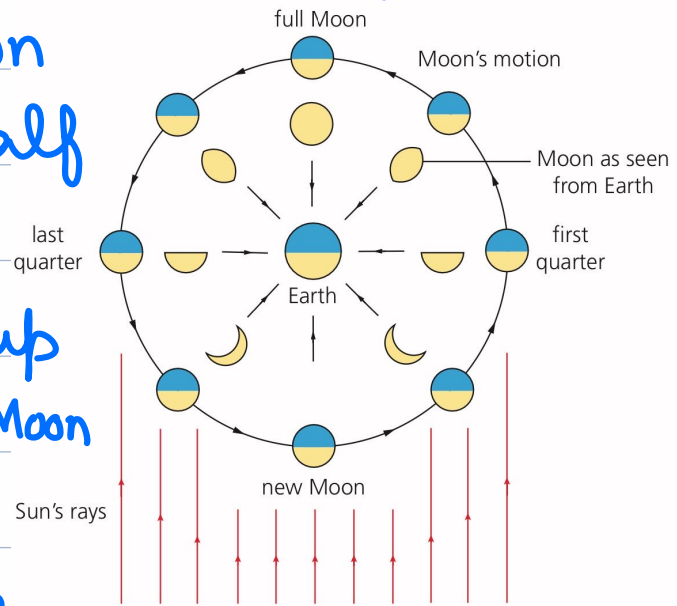
MOON:

Identification: Satellite of Earth

Motion: Circular orbit around Earth about

in 27.5 days (about a month)

Visual impact: Moon reflects light from the Sun. As the Moon orbits the Earth, the half of the Moon that faces the Sun is lit up by the Sun. As the Moon moves, the shape of the light part which can be seen from Earth changes.



Distances and times in the Solar system:

The Earth is approximately 150 million kilometres from the sun. This is similar to circling the Earth 4000 times.

Distances are often expressed in terms of how long it takes light to travel; one light year is the distance travelled by light in a year.

Light from sun takes 500 seconds (approx. eight and a half minutes) To reach at the surface of Earth.

Q.1) Calculate the time for light from the Sun to travel the 150 000 000 km to Earth. Give your answer in minutes.

$$\text{Speed} = \frac{\text{Distance moved}}{\text{time taken}}$$

$$3.0 \times 10^8 = \frac{150\,000\,000\,000}{t} \Rightarrow t = 500\text{ s}$$

$$\text{time} = \frac{500}{60} = 8.33 \text{ minutes.}$$

Orbital speed:

Def. Distance travelled by a planet around sun (or by a moon around a planet) per unit time.

Formula:

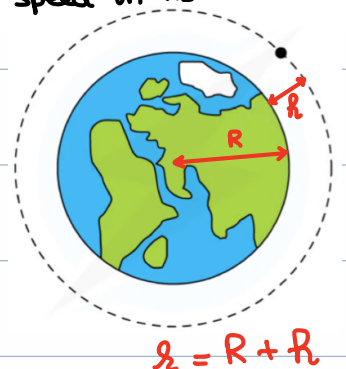
$$\begin{aligned} \text{Distance moved} &= \text{Circumference of orbit} \\ &= 2\pi r \quad r - \text{orbital radius} \end{aligned}$$

Time taken to complete one orbit = Time period (T)

$$\text{So, Speed} = \frac{\text{Distance}}{\text{Time}} \Rightarrow v = \frac{2\pi r}{T}$$

Q.2) The Hubble Space Telescope moves in a circular orbit. Its distance above the Earth's surface is 560 km and the radius of the Earth is 6400 km. It completes one orbit in 96 minutes. Calculate orbital speed in ms^{-1} .

$$\begin{aligned} v &= \frac{2\pi r}{T} \\ &= \frac{2(3.14)(6400 + 560)10^3}{96 \times 60} \\ &= 7.59 \times 10^3 \text{ ms}^{-1} \end{aligned}$$



Q.3) The orbital speed of Earth around the Sun is 30 km / s. Use this value to calculate the distance of Earth from the Sun.

Show your working. State the unit of your answer.

$$v = \frac{2\pi r}{T} \Rightarrow 30 \times 10^3 = \frac{2(3.14)(r)}{365 \times 24 \times 60 \times 60}$$

$$r = 1.51 \times 10^{11} \text{ m}$$

distance = 1.51×10^{11} unit = **m** [3]

Q.4) Light from the Sun travels to the Earth at a speed of 3.0×10^8 m/s.

The time for the complete journey is 500 s.

$$\text{Speed} = \frac{\text{Distance}}{\text{time}} \Rightarrow 3.0 \times 10^8 = \frac{d}{500}$$
$$d = 1.5 \times 10^{11} \text{ m}$$

What is the average orbital speed of the Earth in its orbit around the Sun?

A 60 m/s

B 4.8×10^3 m/s

C 3.0×10^4 m/s

D 1.8×10^6 m/s

$$v = \frac{2\pi r}{T} \Rightarrow v = \frac{2\pi d}{T}$$
$$v = \frac{2(3.14)(1.5 \times 10^{11})}{365 \times 24 \times 60 \times 60} = 2.98 \times 10^4 \text{ m s}^{-1}$$

6.1.2 The Solar System

1 Describe the Solar System as containing:

(a) one star, the Sun

(b) the eight named planets and know their order from the Sun

(c) minor planets that orbit the Sun, including dwarf planets such as Pluto and asteroids in the asteroid belt

(d) moons, that orbit the planets

(e) smaller Solar System bodies, including comets and natural satellites

2 Analyse and interpret planetary data about orbital distance, orbital period, density, surface temperature and uniform gravitational field strength at the planet's surface

3 Know that the strength of the gravitational field:

(a) at the surface of a planet depends on the mass of the planet

(b) around a planet decreases as the distance from the planet increases

4 Know that the Sun contains most of the mass of the Solar System and that the strength of the gravitational field at the surface of the Sun is greater than the strength of the gravitational field at the surface of the planets

5 Know that the force that keeps an object in orbit around the Sun is the gravitational attraction of the Sun

6 Know that the strength of the Sun's gravitational field decreases and that the orbital speeds of the planets decrease as the distance from the Sun increases

SOLAR SYSTEM:

Solar system consists of

(a) a star - the Sun:

Position: Center of Solar system

Mass: Most massive object i.e its mass is about 99.8% of the mass of the solar system.

(b) Eight planets: — Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune

They all orbit around the sun in their fixed elliptical paths depending upon their relative mass.

Inner Planets

(Mercury, Venus, Earth and Mars)

They are small, similar in size, solid and rocky with high density.

Outer planets

(Jupiter, Saturn, Uranus and Neptune)

They are much larger and colder and consists of gases with low density

(c) Minor planets: Pluto, Eris, Ceres, Makemake and Haumea

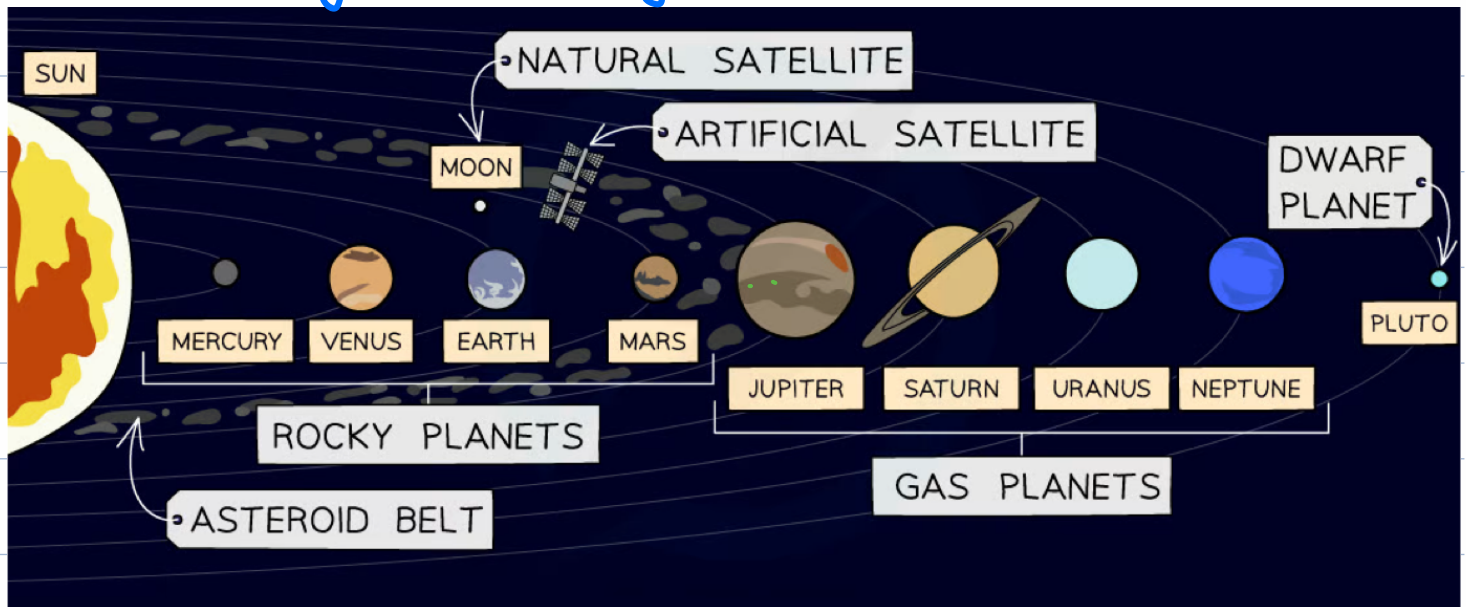
They are actually smaller in size (less than Earth's moon) and mass, so lacks in Gravitational force needed to pull all materials in their orbit.

Pluto orbits around sun at a greater radius than Neptune (TNO → Trans-Neptune objects. It has low density and is composed of rock and ice.

Asteroids: These are the pieces of rocks of different sizes and orbit the sun primarily

between the orbits of Mars and Jupiter in a nearly flat ring called the asteroid belt.

Comets: They consists of frozen gases, rocks, ice and dust left over from the formation of solar system and orbit about the sun. Their density is similar to the outer planets and they follow highly elliptical orbits around sun other than planets. When a comet come closer to sun, it heats up and spews dust and gases into a giant glowing head larger than most planets.



Moon: This is a type of natural satellite that orbit the planets.

Artificial satellites:- Human build these satellites and are generally orbit the Earth. Geostationary → Equator
polar satellite → Poles

Data for the Sun, Earth and Moon

	Mass/kg	Radius/m	Density kg/m ³	Surface gravity /N/kg	Orbital period
Sun	2.0×10^{30}	7.0×10^8	1410	274	
Earth	6.0×10^{24}	6.4×10^6	5520	9.8	365 days (around Sun)
Moon	7.4×10^{22}	1.7×10^6	3340	1.7	27 days (around Earth)

Data for the planets

Planet	Av distance from Sun /million km ↑	Orbit time round sun /days or years ↑	Surface temperature /°C ↓	Density /kg/m ³	Diameter /10 ³ km	Mass /10 ²⁴ kg	Surface gravity /N/kg	No. of moons
Mercury	57.9	88d	350	5427	4.8	0.330	3.7	0
Venus	108.2	225d	460	5243	12.1	4.87	8.9	0
Earth	149.6	365d	20	5514	12.8	5.97	9.8	1
Mars	227.9	687d	-23	3933	6.8	0.642	3.7	2
Jupiter	778.6	11.9y	-120	1326	143	1898	23.1	79
Saturn	1433.5	29.5y	-180	687	120	568	9.0	82
Uranus	2872.5	84y	-210	1271	51	86.8	8.7	27
Neptune	4495.1	165y	-220	1638	50	102	11.0	14

Gravitational Field strength: (g)

$$W = mg \Rightarrow g = \frac{W}{m} \Rightarrow g \propto \frac{M}{d^2}$$

Mass of planet
Distance from center of planet

Nature: Attractive force/field

Def. Attractive pull acting per unit mass of an object.

Dependence:

(i) Mass of planet: $g \propto M$

(ii) Distance from center of planet: $g \propto \frac{1}{d^2}$ i.e.

g decreases when distance from center of planet increases

Example:

At the surface of	$g/\text{N kg}^{-1}$
Sun	274
Earth	9.81
Moon	1.67

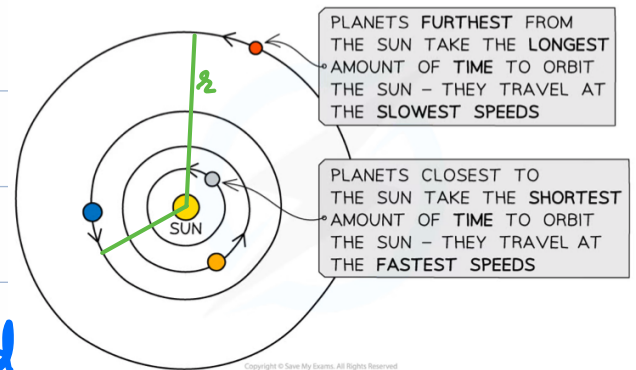
Significance of g in solar system:

1 - Sun is the largest and massive body in the solar system. More than 99% of

the total mass of solar system is concentrated in the sun. Because of heavy mass, Sun exerts a very large gravitational attraction to keep planets, asteroids, comets and other object in a circular orbit around it.

2- As the distance from the sun

increases, gravitational field strength decreases so centripetal force and hence orbital speed



also decreases. This increases the time period of planet.

$$F_g \downarrow, F_c \downarrow, v \downarrow$$

$$F_c = F_g$$

3- An object possess two types of energy during revolution around sun e.g Kinetic energy and potential energy. At closest to the sun, object have maximum Kinetic energy and minimum gravitational potential energy.

Q.5) The average distance of Mercury from the sun is 58 million km. Calculate

(1) the distance travelled by Mercury in one orbit of the sun.

$$\begin{aligned} \text{distance travelled} &= \text{circumference} = 2\pi r = 2(3.14)(58 \times 10^6 \times 10^3) \\ &= 3.64 \times 10^{11} \text{ m} \end{aligned}$$

(ii) the orbital speed of Mercury if it takes 88 days to complete one orbit.

$$v = \frac{\text{distance}}{\text{time}} = \frac{2\pi r}{T} = \frac{3.64 \times 10^{11}}{88 \times 24 \times 60 \times 60}$$

$$v = 4.79 \times 10^4 \text{ m s}^{-1}$$

Q.6.) A comet enters the Solar system from beyond Pluto and approaches the Sun.

(a) State how the gravitational field strength changes as the distance from the Sun decreases.

Increases as $g \propto \frac{1}{(\text{Distance from Sun})^2}$

(b) State the change in speed of the comet as it approaches the Sun.

(Speed) $\uparrow$ as $g \uparrow$ so Force (weight) $\uparrow$
Hence (K.E) $\uparrow$ and (P.E) $\downarrow$

Q.7)

Orbiting Objects or Bodies in Our Solar System

Body or Object	What it Orbits
Planet	Sun
Moon	Planet
Comet	Sun
Asteroid	Sun
Artificial satellite	Any object or body in solar system

Stars and the Universe

6.2 Stars and the Universe

6.2.1 The Sun as a star

- 1 Know that the Sun is a star of medium size, consisting mostly of hydrogen and helium, and that it radiates most of its energy in the infrared, visible and ultraviolet regions of the electromagnetic spectrum
- 2 Know that stars are powered by nuclear reactions that release energy and that in stable stars the nuclear reactions involve the fusion of hydrogen into helium

The Sun:

The Sun is a stable medium-sized star and is made up of about 75% hydrogen and about 24% helium. The rest 1% is made up of other elements, such as oxygen and carbon.

The thermo-nuclear reactions in the Sun radiates energy. About 40% of this energy is visible light, about 50% is infrared radiations, and the remainin 10% is ultraviolet.

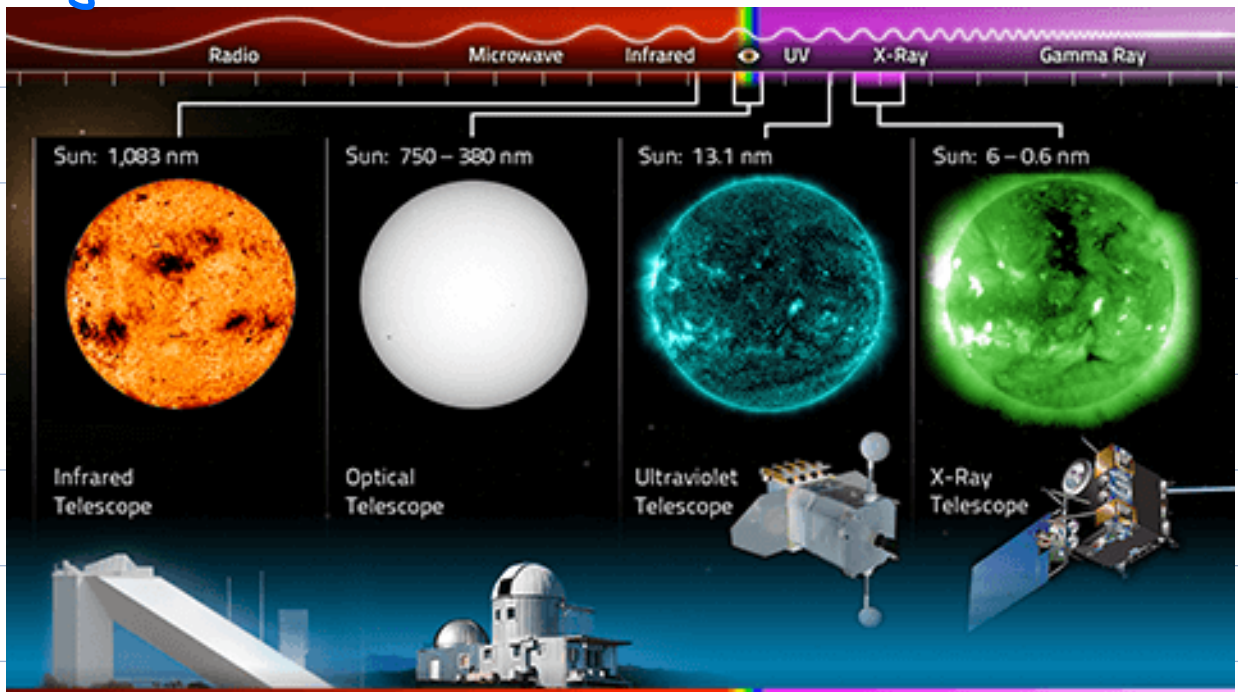
Nuclear reactions in stars: ${}^2_1\text{H} + {}^3_1\text{H} \longrightarrow {}^4_2\text{He} + {}^1_0\text{n} + Q$

Stars releases energy due to thermo-nuclear Fusion of hydrogen into helium. The core temperature of Sun is $1.50 \times 10^7 \text{K}$ and this temperature is sufficient enough to sustain Nuclear Fusion reaction.

Since, energy of star is spread over a big surface, so its temperature is lower at the surface of Sun (about 5800K).

Colour of Sun:-

The colour of sun is white. It looks yellow due to Earth's atmosphere because beam of sunlight is scattered and dispersed while passing through different medium.



*The Sun as seen in different colors by different telescopes.
Image credit: The University of Chicago.*

6.2.2 Stars

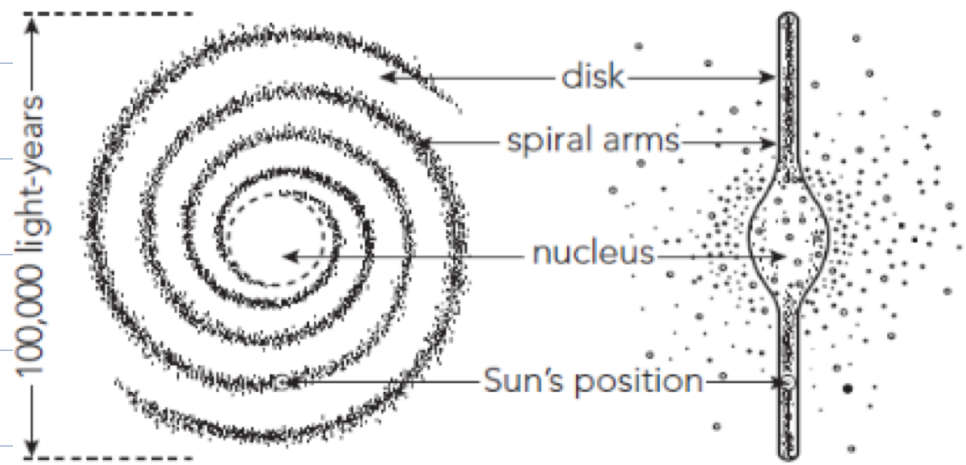
1 State that:

- (a) galaxies are each made up of many billions of stars
- (b) the Sun is a star in the galaxy known as the Milky Way
- (c) other stars that make up the Milky Way are much further away from the Earth than the Sun is from the Earth
- (d) astronomical distances can be measured in light-years, where one light-year is the distance travelled in a vacuum by light in one year

Galaxies: The force of gravity pulls stars together in groups called galaxies. A galaxy is a large collection of stars; there are billions of stars in a galaxy.

Composition: Apart from stars, galaxies consist of clouds of gas, mostly hydrogen and dust.

Milky Way: It is the spiral galaxy to which our solar system belongs. There might be 200 billion (2×10^{11}) stars in the Milky way. It is spinning and it takes our Solar system about 225 million years to travel once around the galaxy.



We are near the outer edge of the galaxy, so what we see when we look at the Milky way is the galaxy's central bulge.

The Sun is a star in this galaxy and other stars in the Milky way are further away from Earth than the Sun is from the Earth.

Galaxies travel in groups. The nearest spiral galaxy is the Andromeda Galaxy.

Astronomical distances:

Astronomers have many techniques to measure distance in space. For example, when you look into the night sky, the light you see from the stars has been travelling

for many years. Astronomers use this idea as a way of measuring vast distances.

A **light year** is a measure of distance (not time). It is the distance travelled by light through space in one year.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$3.0 \times 10^8 = \frac{d}{365 \times 24 \times 60 \times 60} \Rightarrow d = 9.5 \times 10^{15} \text{ m}$$

Q7.) The Universe is composed of stars and galaxies. Which of the following statements are false?

A → The Universe is a collection of galaxies.

B → There are billions of stars in a galaxy.

C → The Sun is a star

D → Our Solar system belongs to the Andromeda galaxy. (Our Solar system belongs to Milky way)

Q.8) What is a light year?

A → the distance light travels in one year.

B → 366 days

C → the time it takes to travel in one year.

D → 1.44×10^{11} metres $d = 9.5 \times 10^{15} \text{ m}$

Q.9) How long does it take light to reach the Earth from a galaxy 10 million light years distance from Earth?

$$d = vt$$

$$d = (3 \times 10^8)$$

$$(365 \times 24 \times 60 \times 60)$$

$$d = 9.5 \times 10^{15} \text{ m}$$

A 10 years

B 30 years

C 300 thousand years D 10 million years

$$\text{Distance in 10 million light years} = (10 \times 10^6) (9.5 \times 10^{15}) = 9.5 \times 10^{22} \text{ m}$$

Q.10. Why do stars shine?

A → They are burning.

B → Nuclear Fusion is taking place inside it.

C → They are made of hot gases that are heated up when they collapsed from gas clouds.

D → They are made of ether.

$$d = vt$$

$$9.5 \times 10^{22} = (3 \times 10^8) t$$

$$t = \frac{9.5 \times 10^{22}}{3 \times 10^8} = 3.17 \times 10^{14} \text{ s} = \frac{3.17 \times 10^{14}}{365 \times 24 \times 60 \times 60} \text{ years}$$

Q.11) In the Sun, energy is transferred from an energy store.

Which energy is being released and what is the name of the reaction that releases this energy inside the Sun?

	energy	reaction
A	chemical	burning
B	chemical	radiation
C	nuclear	fission
D	nuclear	fusion

Life cycle of a star:

2 Describe the life cycle of a star:

- a star is formed from interstellar clouds of gas and dust that contain hydrogen
- a protostar is an interstellar cloud collapsing and increasing in temperature as a result of its internal gravitational attraction
- a protostar becomes a stable star when the inward force of gravitational attraction is balanced by an outward force due to the high temperature in the centre of the star
- all stars eventually run out of hydrogen as fuel for the nuclear reaction
- most stars expand to form red giants and more massive stars expand to form red supergiants when most of the hydrogen in the centre of the star has been converted to helium
- a red giant from a less massive star forms a planetary nebula with a white dwarf at its centre
- a red supergiant explodes as a supernova, forming a nebula containing hydrogen and new heavier elements, leaving behind a neutron star or a black hole at its centre
- the nebula from a supernova may form new stars with orbiting planets

The Solar System and Orbits

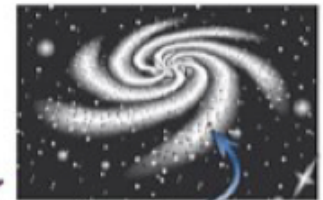
The Sun is the centre of our solar system. It's orbited by eight planets, along with a bunch of other objects.

Our Solar System has **One Star** — The **Sun**

The solar system is all the stuff that orbits our Sun. This includes things like:

- 1) Planets — these are large objects that orbit a star. There are eight in our solar system. They have to be large enough to have "cleared their neighbourhoods". This means that their gravity is strong enough to have pulled in any nearby objects apart from their natural satellites.
- 2) Dwarf planets, like our pal Pluto. These are planet-like objects that orbit stars, but don't meet all of the rules for being a planet.
- 3) Moons — these orbit planets. They're a type of natural satellite (i.e. they're not man-made).
- 4) Artificial satellites are satellites that humans have built. They generally orbit the Earth.

Our solar system is a tiny part of the Milky Way galaxy. This is a massive collection of billions of stars that are all held together by gravity.



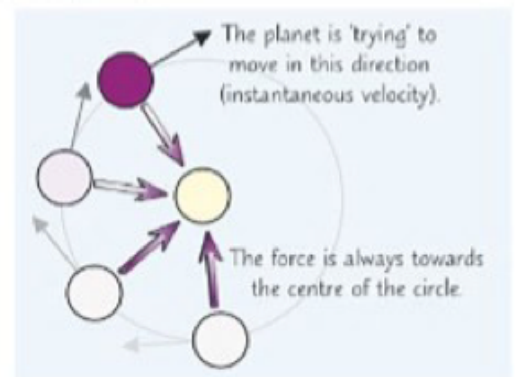
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The planets are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

Gravity Provides the **Force** That Creates Orbits

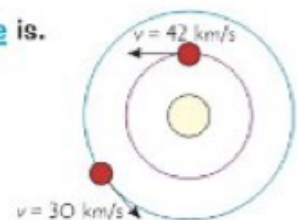
- 1) The planets move around the Sun in almost circular orbits (the same goes for the Moon orbiting the Earth).
- 2) If an object is travelling in a circle it is constantly changing direction, which means it is constantly accelerating. (Just like a car going around a roundabout, pages 60-61.)
- 3) This also means it is constantly changing velocity (but **NOT** changing speed).
- 4) For an object to accelerate, there must be a force acting on it (p.64). This force is directed towards the centre of the circle.
- 5) This force would cause the object to just fall towards whatever it was orbiting, but as the object is already moving, it just causes it to change its direction.
- 6) The object keeps accelerating towards what it's orbiting but the instantaneous velocity (which is at a right angle to the acceleration) keeps it travelling in a circle.
- 7) The force that makes this happen is provided by the gravitational force (gravity) between the planet and the Sun (or between the planet and its satellites).

A satellite is an object that orbits a second, more massive object.



The **Size** of the Orbit Depends on the Object's **Speed**

- 1) The closer you get to a star or planet, the stronger the gravitational force is.
- 2) The stronger the force, the faster the orbiting object needs to travel to remain in orbit (to not crash into the object that it's orbiting).
- 3) For an object in a stable orbit, if the speed of the object changes, the size (radius) of its orbit must do so too. Faster moving objects will move in a stable orbit with a smaller radius than slower moving ones.



Origin of stars:

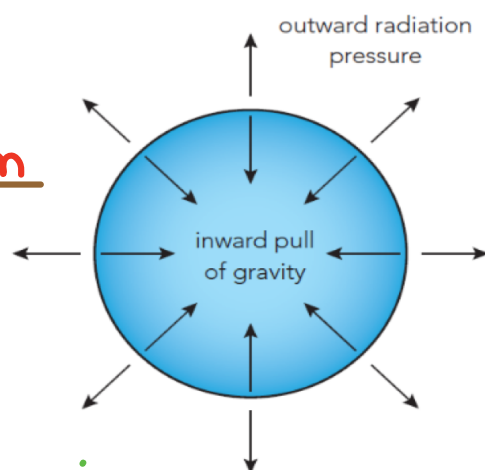
When interstellar (interplanetary) clouds of dust and gas containing hydrogen also called **nebula** experience a strong Gravitational force due to their masses and form a hot ball of gas, known as **protostar**. As the mass of the protostar increases, its gravitational force increases which results in a increase in temperature and gravitational potential energy changes to kinetic energy.

When the **protostar** is hot enough, Nuclear Fusion is initiated and a **star** is born. If the young star has a very large mass, it forms a blue or white star. If it has a smaller mass, such as our Sun, it forms a yellow or red dwarf. Life cycle of a star depends upon its mass.

Life cycle of stars:

(a) Stable star - the Sun

During the main sequence, the star is in equilibrium



A star is stable when the inward pull of gravity is balanced by the outward push of radiation pressure, which is actually a force.

and is said to be stable.

The force of gravity pulling it inward are balanced by the opposing forces acting outward caused due to thermal pressure of radiations. Now the star is balanced and may last 10 000 million (10^9) years.

(b) Unstable star - Red Giant

After 10^9 years, hydrogen causing the Fusion reactions in the star will begin to run out. Now there is less energy being produced by Nuclear Fusion to sustain the outward thermal pressure and the core collapses inward under gravitational attraction. This also increases the kinetic energy of particles to initiate other series of Fusion reactions. These reactions causes the outer part of star to expand and subsequently cooling of gases and the surface and looks like red.

If the star has mass nearly equal to that of our sun then it is red giant. A star much larger than sun will become red super giant.

(c) i) Red Giant Star:

Once this second stage of Fusion has finished, the star will become unstable and eject the outer layer of dust and gas also called

a **planetary nebula**. The remaining core will collapse completely, due to pull of gravity, and the star will become **white dwarf**. The white dwarf will be cooling down and therefore, the amount of heat radiated by it decreases. When the star has lost significant amount of energy, it becomes a **black dwarf**. It will continue to cool until it eventually **disappears** from sight.

(c)(ii) Red Super Giants:

Once the fusion inside the red super Giant finish, the core of star will collapse suddenly causing a gigantic explosion. This is called **supernova**. This provides the energy required to create elements heavier than iron and push them into space as a nebula along with lighter elements. The nebula provides the building blocks for possible future stars and solar system. If the mass of remaining core is less, a **neutron star** is formed and if the mass of core is large, it collapse until it becomes so dense that not even light can escape and the star becomes a **black hole**.

The Life Cycle of Stars



The Universe

6.2.3 The Universe

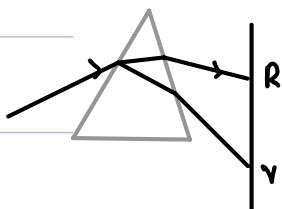
- 1 Know that the Milky Way is one of many billions of galaxies making up the Universe and that the diameter of the Milky Way is approximately 100 000 light-years
- 2 Describe redshift as an increase in the observed wavelength of electromagnetic radiation emitted from receding stars and galaxies
- 3 Know that the light from distant galaxies shows redshift and that the further away the galaxy, the greater the observed redshift and the faster the galaxy's speed away from the Earth
- 4 Describe, qualitatively, how redshift provides evidence for the Big Bang theory

The Universe:

The Milky way is one of the billions of galaxies that make up the Universe. The diameter of the Milky way galaxy is around 100 000 light-years and it contains 800 billion or more stars.

Redshift:

Astronomers observe the light from distant galaxies using powerful telescopes and came to this conclusion that all observed wavelengths of all spectral lines are longer than the ones observed in the laboratory. This shows that they all are moving away from us.



Absorption lines from the Sun



Absorption lines from a supercluster of galaxies BAS11

$v = 0.07c$, $d = 1$ billion light years



The absorption lines in the spectrum of the galaxies are all shifted to longer wavelengths - redshifted. The top spectrum is the spectrum from a 'stationary source', the Sun.

Note:

- 1- The observed wavelength of electromagnetic wave is longer for a receding source and shorter for an approaching source.

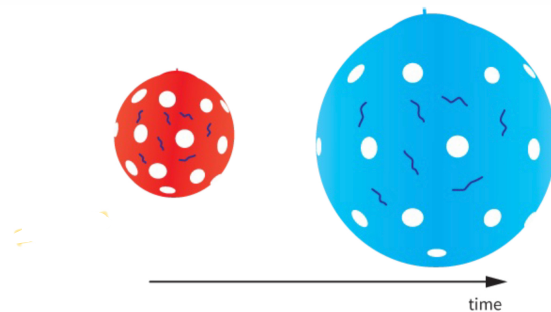
- 2- Light from distant galaxies shows redshift and that the further away the galaxy, the greater the observed redshift and the faster the galaxy's speed away from the Earth.
- 3- The redshift in the light from distant galaxies provides evidence that the Universe is expanding and gives support to the Big Bang theory of the formation of the Universe.

The Big Bang Theory:

If all the galaxies are receding from each other, it follows that in the past they must have been closer together. It is therefore possible that initially all the matter in the Universe was packed together in an extremely dense state. An

explosion by Big Bang theory split masses apart and

they are moving away from each other like dots shown on a balloon which is being inflated.



The galaxies are modelled as dots on the surface of a balloon. Expansion of the balloon makes all the dots move **away** from each other.

Red-shift and the Big Bang

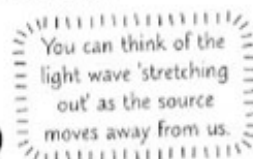
'How it all began' is a tricky question that we just can't answer. Our best guess at the minute is the Big Bang.

The Universe Seems to Be *Expanding*

As big as the universe already is, it looks like it's getting even bigger.

All its galaxies seem to be moving away from each other. There's good evidence for this...

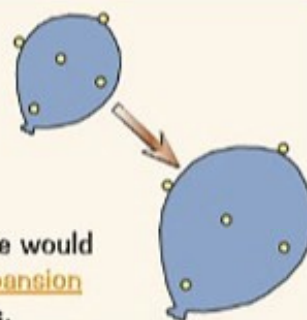
- 1) When we look at light from most distant galaxies, we find that the wavelength has increased.
- 2) The wavelengths are all longer than they should be — they're shifted towards the red end of the spectrum. This is called red-shift.
- 3) This suggests the source of the light is moving away from us. Measurements of the red-shift indicate that these distant galaxies are moving away from us (receding) very quickly — and it's the same result whichever direction you look in.
- 4) More distant galaxies have greater red-shifts than nearer ones. This means that more distant galaxies are moving away faster than nearer ones.
- 5) The inescapable conclusion appears to be that the whole universe (space itself) is expanding.



Imagine a balloon covered with pompoms.
As you blow into the balloon, it stretches.
The pompoms move further away from each other.

The balloon represents the universe and each pompom is a galaxy. As time goes on, space stretches and expands, moving the galaxies away from each other.

This is a simple model (balloons only stretch so far, and there would be galaxies 'inside' the balloon too) but it shows how the expansion of space makes it look like galaxies are moving away from us.



This Evidence Suggests the Universe *Started with a Bang*

So all the galaxies are moving away from each other at great speed — suggesting something must have got them going. That 'something' was probably a big explosion — the Big Bang. Here's the theory...

- 1) Initially, all the matter in the universe occupied a very small space. This tiny space was very dense (p.38) and so was very hot.
- 2) Then it 'exploded' — space started expanding, and the expansion is still going on.



Revision Questions

Topic — short, sweet and super interesting. Now it's time to check you filled all that space in your head.

- Try these questions and tick off each one when you get it right.
- When you've done all the questions under a heading and are completely happy with it, tick it off.

The Life Cycle of Stars

☐

- 1) What is a nebula?
- 2) What causes the rise in temperature that leads to nuclear fusion in a protostar?
- 3) What causes a main sequence star to remain stable for a long time?
- 4) What happens to a star about the same size as our Sun when it begins to run out of hydrogen?
- 5) What is a black dwarf and how is it made?
- 6) At what stage in a star's life cycle are elements heavier than iron formed?
- 7) True or false? The Sun will eventually turn into a black hole.

☐
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☐

The Solar System and Orbits

☐

- 8) How many stars are there in our solar system?
- 9) What do planets and dwarf planets orbit?
- 10) True or false? The moon is an artificial satellite.
- 11) What galaxy is our solar system part of?
- 12) What is the approximate shape of the planets' orbits around the Sun?
- 13) True or false? An object in a stable orbit has a continually changing speed.
- 14) What is the name of the force that pulls an orbiting object towards Earth?
- 15) State the direction of the orbiting object's instantaneous velocity in relation to this force.
- 16) How does the strength of a planet's gravitational force change as you get closer to its surface?
- 17) What happens to the speed of an orbiting object if its orbital radius increases?
Explain why this happens.

☐
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Red-shift and the Big Bang

☐

- 18) What is red-shift?
- 19) True or false? Very distant galaxies are moving away faster than ones closer to us.
- 20) Give two limitations to the balloon model of the expanding universe.
- 21) Briefly describe the ideas that make up the Big Bang theory.
- 22) What did scientists discover about the movement of galaxies in 1998 and how did they discover this?
- 23) True or false? New evidence that disproves a popular theory is ignored.

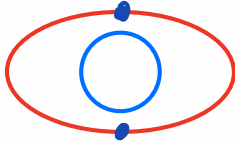
☐
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☐

Name _____

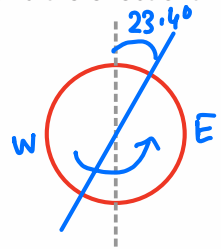
Class _____

Space Physics

1. Which line in the table best describes the angle of the Earth's axis from the vertical, and the effect of this tilt?



	Angle of tilt	Effect of tilt
☒ A	23.4°	Rising and setting of the Sun
☐ B	23.4°	Changing of the seasons
☐ C	66.6°	Rising and setting of the Sun
☐ D	66.6°	Changing of the seasons



2. Which line in the table correctly identifies the length of one year, one day and daylight hours during an equinox, all as experienced on Earth?

Day and Night are of equal hours duration and Sun is above the equator. The equinox happens twice in a year eg in sep. and March.

	One Earth year / days	One Earth day / hours	Day light hours during an equinox / hours
A	365 ✓	12 X	24 X
B	365 ✓	12	24
C	365	24	12
D	366	24	12

3. The orbit of the Earth around the Sun can be thought of as following a circular path with a circumference of 9.42×10^8 km. What is the approximate orbital speed of the Earth?

$$V = \frac{2\pi r}{T} = \frac{2(3.14)(9.42 \times 10^8)}{365 \times 24 \times 60 \times 60} = 100\,000 \text{ km/h}$$

A 10 000 km/h

☒ B 100 000 km/hC 2.5×10^6 km/hD 50×10^6 km/h

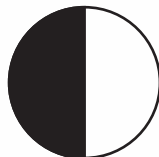
4. Which of the following objects are described as orbiting the Sun?

1. Asteroids ✓ 2. Comets ✓ 3. Galaxies X 4. Moons X

☒ A 1 and 2 only☐ B 1,2 and 3☐ C 1,2, and 4☐ D 1,2,3 and 4

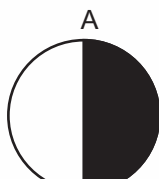
5. The diagram shows a phase of the Moon.

Full
Quarter
Moon

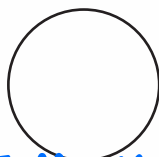


Which of the four options shows the Moon 7 days later?

A
Last Quarter Moon



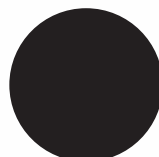
B
Full Moon



C
Crescent moon after 7 day



D
New Moon



6. Which of the planets listed is the gaseous planet which is closest to the Sun?

(A) Jupiter *Gas planet* B Mars *Rocky planet* C Saturn *Gas planet* D Uranus *Gas planet*

7. The Earth and Moon are held together by a gravitational force acting between them.

Select the line which describes the change to the gravitational force if the mass of the Earth was double its actual value, and if the distance between the two bodies was twice the actual value.

Effect on gravitational force if

$$F \propto g \propto \frac{M}{d^2}$$

	Mass of Earth doubles	Earth-Moon distance doubles
(A)	Double ✓	Drop to one quarter ✓
B	Stay the same	Stay the same
C	Double ✓	halve
D	Drop to one quarter	Double

$$F \propto g \propto M, g \rightarrow \text{double}$$

$$F \propto g \propto \frac{1}{d^2}, g \rightarrow \text{Quarter}$$

8. The orbit of Venus around the Sun is treated as a circular path with an average radius of 108.2 million km.

The orbital period of Venus is 225 days.

$$v = \frac{2\pi r}{T} = \frac{2(3.14)(108.2 \times 10^6)}{225 \times 24}$$

What is the orbital speed of Venus?

- A 40 000 km/h B 70 000 km/h C 126 000 km/h D 810 000 km/h

9. Which two elements are the most commonly found in the Sun?

A Hydrogen and oxygen ✓
 B Helium and carbon ✓
 (C) Hydrogen and helium ✓
 D Nitrogen and oxygen ✓

10. Which is the best description of redshift?

A Distant galaxies only emit light from the red end of the spectrum.
 B Only red light can reach Earth from very distant stars.
 C Stars in the Milky Way mostly emit light at the red end of the spectrum,
 (D) The light received from distant galaxies is moved towards the red end of the spectrum.

11. Light from distant galaxies appears red-shifted to observers on Earth.

Which of the following ideas does this provide evidence for?

1. The Big Bang Theory ✓
 2. The Universe is expanding ✓
 3. Distant Galaxies are moving both towards us and away from us ✓

- A 1 only (B) 1 and 2 C 2 only D 1, 2 and 3

12. How many years would it take light to travel from the centre of the Milky Way to the outer edge?

A 500 years B 50 000 years C 100 000 years (D) 100 000 light years

Diameter of Milky Way Galaxy is 100 000 light-years. So light take 100 000 year to travel across it

13. What is the process which makes stable stars such as the Sun shine, and which element is produced in this reaction?

	Process	Element
A	fusion	oxygen
B	fission	hydrogen
(C)	fusion	helium
D	fission	helium

$$v = \frac{d}{t} \Rightarrow d = v t$$

$$= (3 \times 10^8) (365 \times 24 \times 60 \times 60) = 9.46 \times 10^{15} \text{ m}$$

14. Which is the correct value of one light year? *distance travelled by light in one year*

A $95 \times 10^{15} \text{ km}$ (B) $9.5 \times 10^{15} \text{ m}$ C $95 \times 10^{15} \text{ s}$ D $9.5 \times 10^{15} \text{ days}$

15. The planet Uranus has an average orbital radius of 4 500 million km. How long does it take light from the Sun to travel to Uranus?

A 4 minutes B 1 hour (C) 4 hours D 4 days

$$v = \frac{d}{t} \Rightarrow t = \frac{d}{v} = \frac{4500 \times 10^6 \times 10^3}{3.0 \times 10^8} \Rightarrow t = \frac{?}{3600} = 4 \text{ h}$$

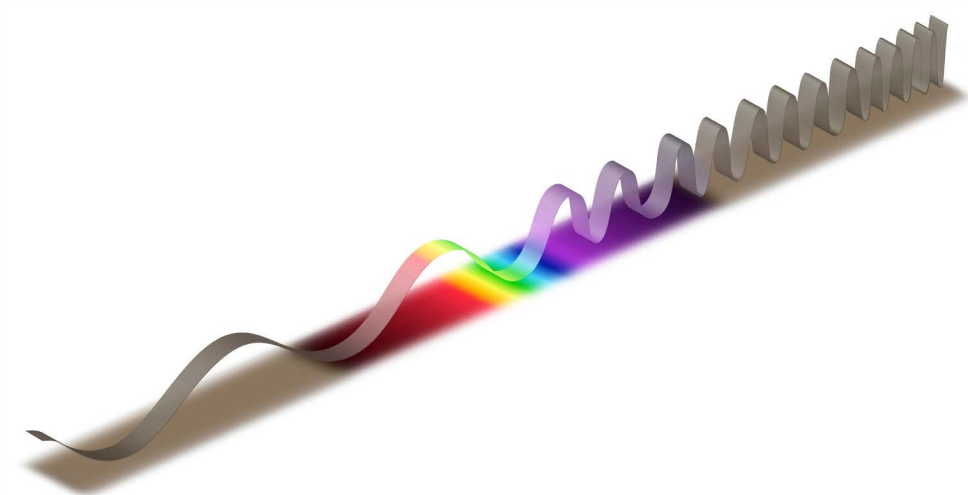


Space Physics Practice Questions

Cambridge O Level

Physics 5054

To accompany the revised syllabus for examination from 2023.



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Introduction

The purpose of this booklet is to provide additional practice questions and answers for the Space Physics topic which has been introduced to Cambridge O Level Physics (5054) for first assessment in 2023.

Practice questions have been provided to exemplify a range of types of questions which could appear in examinations.

The answers and a typical mark scheme are also provided.

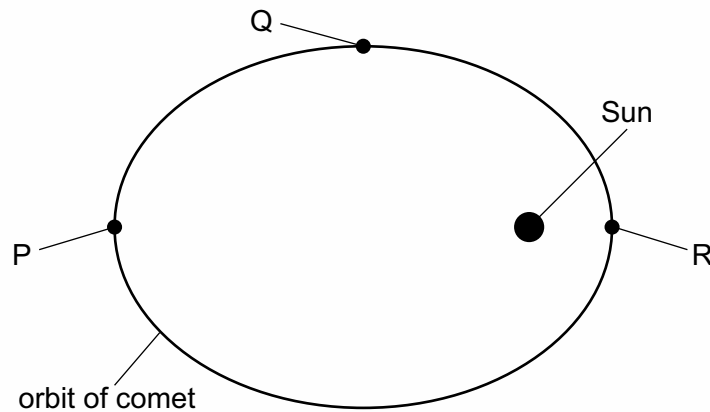
Other support materials are available on the School Support Hub www.cambridgeinternational.org/support

Paper 1 example questions

- 1 Which row in the table correctly lists a galaxy, a planet and a star?

	galaxy	planet	star
A	Milky Way	Mars	Moon
B	Milky Way	Earth	Sun
C	Solar System	Mars	Sun
D	Solar System	Earth	Moon

- 2 The orbit around the Sun of a comet is not circular. The distance between the Sun and the comet varies, as shown in the diagram.

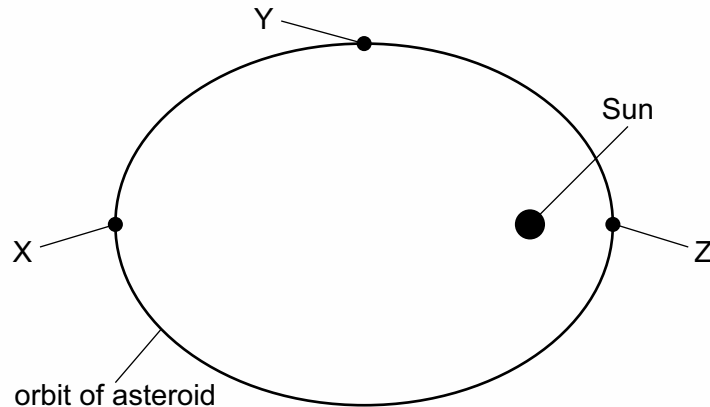


Where, in the orbit, is the speed of the comet the greatest and where is it the smallest?

	speed greatest at	speed smallest at
A	P	R
B	P and R	Q
C	Q	P and R
D	R	P

- 3 The orbit around the Sun of a particular asteroid (dwarf planet) is not circular. The distance between the Sun and the asteroid varies, as shown in the diagram.

The asteroid possesses both gravitational potential energy (GPE) and kinetic energy (KE).



Which energy transfer takes place as the asteroid moves from X to Y and as it moves from Y to Z?

	moving from X to Y	moving from Y to Z
A	GPE to KE	GPE to KE
B	GPE to KE	KE to GPE
C	KE to GPE	GPE to KE
D	KE to GPE	KE to GPE

- 4 Ceres and Vesta are two asteroids (dwarf planets) that orbit the Sun. The orbits of Ceres and Vesta are approximately circular.

The Sun's gravitational field at the orbit of Ceres is weaker than that at the orbit of Vesta.

How does the orbital speed of Ceres and its orbital period compare with the orbital speed and orbital period of Vesta?

	orbital speed of Ceres	orbital period of Ceres
A	smaller than that of Vesta	smaller than that of Vesta
B	smaller than that of Vesta	greater than that of Vesta
C	greater than that of Vesta	smaller than that of Vesta
D	greater than that of Vesta	greater than that of Vesta

- 5 The Sun emits energy in the form of electromagnetic radiation.

Which three components of the electromagnetic spectrum account for almost all of this energy?

- A infrared, microwave and ultraviolet
- B infrared, microwave and visible light
- C infrared, ultraviolet and visible light
- D microwave, ultraviolet and visible light

- 6 When, in the life cycle of a star, are heavy elements produced?

- A when a cloud containing hydrogen collapses
- B when a protostar becomes stable
- C when a red supergiant explodes
- D when a white dwarf is formed

- 7 The statements J, K, L and M describe different stages in the life cycle of a small star.

- J A cloud containing hydrogen collapses due to gravitational attraction.
- K A red giant is produced.
- L A white dwarf is produced.
- M Hydrogen undergoes nuclear fusion to produce helium.

What is the order (first to last) in which the stages occur?

- A J, M, K, L
- B J, M, L, K
- C M, J, K, L
- D M, J, L, K

- 8 What provides evidence for the Big Bang Theory?

- A the expulsion of heavy elements into space during a supernova explosion
- B the increase in the observed wavelength of radiation emitted by distant galaxies
- C the nuclear fusion reaction that takes place at the centre of stable stars
- D the smaller orbital speeds of the planets that are further from the Sun

Paper 1 example questions mark scheme

Question	Answer	Marks
1	B	1
2	D	1
3	A	1
4	B	1
5	C	1
6	C	1
7	A	1
8	B	1

Paper 2 example questions

1 The Solar System contains the Sun, planets, moons, asteroids and comets.

(a) State the name of the planet closest to the Sun.

..... [1]

(b) Describe the difference between a planet and a moon.

.....
 [2]

(c) State **two** differences between an asteroid and a comet.

1

 2
 [2]

(d) The planet closest to the Sun orbits the Sun in 88 days.

State **two** reasons why the Earth takes longer to orbit the Sun than this planet.

1
 2 [2]

(e) (i) State the time taken for light to travel from the Sun to the Earth.

..... [1]

(ii) Using your answer to **(e)(i)**, calculate the distance from the Sun to the Earth.

Show your working.

distance = m [2]

[Total: 10]

- 2 (a) A year, a month and a day can be defined in terms of the motions of the Earth, the Moon and the Sun.

Using these motions, complete the following sentences.

- (i) A year is the time for

..... [1]

- (ii) A month is the time for

..... [1]

- (iii) A day is the time for

..... [1]

- (b) Earth, Jupiter and Saturn are three planets in the Solar System.

Table 2.1 gives information about these planets.

Table 2.1

planet	<u>orbital speed</u> km / h	time to orbit Sun / years
Earth	1.1×10^5	1.0
Jupiter	4.7×10^4	12
Saturn	3.5×10^4	29

Using data from Table 2.1, calculate the average radius of Saturn's orbit around the Sun.

Give your answer in km.

radius = km [3]

(c) Fig. 2.1 shows one alignment with the Sun of Earth, Jupiter and Saturn.

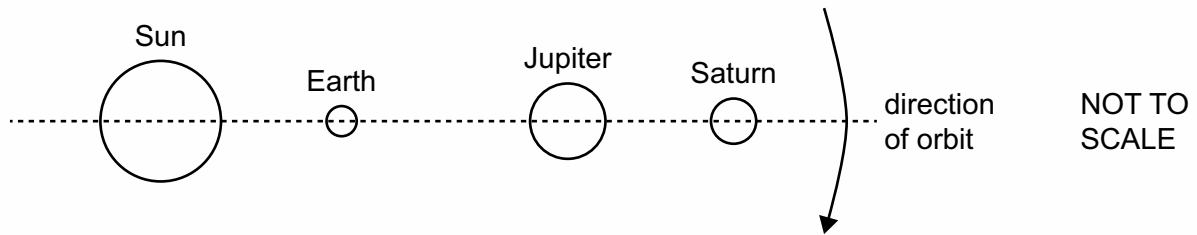


Fig. 2.1

(i) In Fig. 2.1, the three planets and the Sun are in line.

29 years later, Saturn returns to the same position relative to the Sun, but Earth and Jupiter are **not** in the same line.

Use data from Table 2.1 to explain why.

.....

.....

..... [2]

(ii) Complete Fig. 2.2 to show the positions of the three planets exactly 6 years after the diagram in Fig. 2.1.

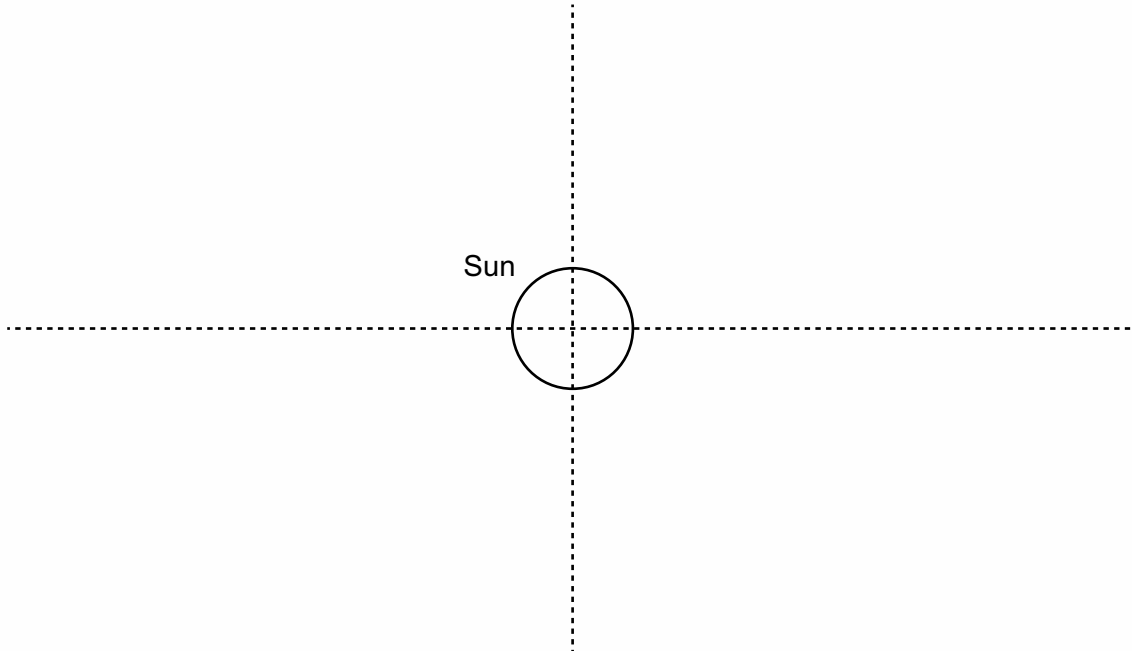


Fig. 2.2

[3]

[Total: 11]

- 3 Table 3.1 lists some of the planets in the Solar System and their average distances from the Sun.

Table 3.1

planet	average distance from the Sun / km
Mercury	5.8×10^7
	1.1×10^8
Earth	1.5×10^8
	2.3×10^8
Jupiter	7.8×10^8

- (a) Complete Table 3.1 by writing in the names of the missing planets. [2]

- (b) As the planets move around the Sun, the distance between Earth and Jupiter changes.

Use Table 3.1 to calculate the smallest distance and the largest distance between Earth and Jupiter.

smallest distance =

largest distance =

[2]

- (c) A planet is in orbit around a distant star. As it passes in front of the star, it blocks some of the light of the star from reaching Earth, reducing the brightness of the star as seen from Earth.

Fig. 3.1 shows the view, from Earth, as the planet passes in front of the star.

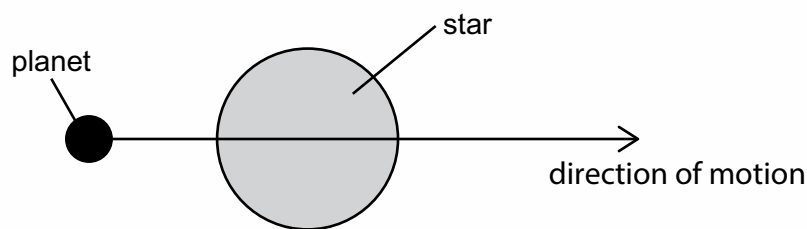


Fig. 3.1

Fig. 3.2 shows an incomplete graph of how the brightness of the star varies with time, as seen from Earth.

The planet is in front of the star from time t_1 to time t_2 .

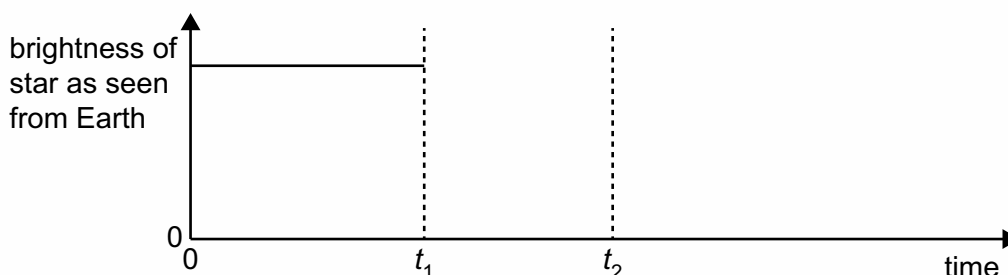


Fig. 3.2

- (i) Complete Fig. 3.2 to show how the brightness of the star varies with time. [2]
- (ii) Describe how the completed Fig. 3.2 should be changed to show a larger planet passing at the same distance in front of the same star.
-
-
- [2]
- (iii) Suggest how an astronomer uses the changes in the brightness of the star to measure the orbital period of the planet round the star.
-
-
- [1]

[Total: 9]

- 4 When astronomers look at distant galaxies, they find that the frequency of the light from the galaxy is shifted towards the red end of the spectrum.

- (a) (i) Apart from visible light, state **two** types of electromagnetic wave that contain most of the energy emitted by a stable star.

..... [1]

- (ii) Describe what is meant by the red end of the spectrum.

.....

..... [1]

- (b) The speed of recession of a galaxy is the speed at which the galaxy is moving away from Earth.

Astronomers use redshift to measure the speed of recession.

Astronomers can also measure the distance d of the galaxy from Earth. Such large distances are measured in light-years.

Astronomers can estimate the time T that the galaxy has been moving away from Earth using the equation

$$T = \frac{d}{\text{speed of recession}}$$

Table 4.1 shows their estimates for four different galaxies.

Table 4.1

d / light-years	<u>speed of recession</u> km/s	T / billions of years
4.9×10^7	1100	13
1.0×10^8	2400	13
3.2×10^8	6700	14
3.9×10^8	9000	13

- (i) State what is meant by a galaxy.

.....

..... [2]

- (ii) Show that 1.0 light-year is equal to 9.5×10^{15} m.

[3]

- (iii) Describe the relationship between the speed of recession of a galaxy and its distance from Earth.

.....
..... [1]

- (iv) Explain how the values of T provide evidence for the Big Bang Theory of the origin of the Universe.

.....
.....
..... [2]

[Total: 10]

5 (a) The Sun produces energy from nuclei of hydrogen. Hydrogen nuclei are positively charged.

(i) Describe the process that produces this energy.

.....

 [2]

(ii) Explain why this process needs a high temperature to occur.

.....

 [2]

(b) (i) Some stars are initially much more massive than the Sun.

Describe the life cycle of such a star, after it has passed through the stage of being a protostar.

.....

 [4]

(ii) Although the early Universe contained only light elements such as hydrogen, it now contains a number of very heavy elements.

State in which part of the life cycle of a star these very heavy elements are formed.

.....
 [1]

[Total: 9]

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Abbreviations and guidance

/	Alternative answers for the same marking point.
<u>underline</u>	Actual word underlined must be used by candidate (grammatical variants accepted).
(brackets)	The word or phrase in brackets is not required but sets the context.
AND / and	Statements on both sides of the AND are needed for that mark.
OR / or	Indicates alternative answers, any one of which is satisfactory for scoring the marks.
NOT / not	Indicates that an incorrect answer is not to be disregarded but cancels another otherwise correct alternative offered by the candidate for this mark.
Accept / A	A less than ideal answer which should be marked correct.
Ignore / Ig	Indicates that something which is not correct or irrelevant is to be disregarded.
e.c.f.	'error carried forward'
o.w.t.t.e.	'or words to that effect'
s.f.	'significant figures' – answers are normally acceptable to any number of significant figures ≥ 2 . Any exceptions to this general rule will be specified in the mark scheme.
Arithmetic errors	If the only error in arriving at a final answer is clearly an arithmetic one, all but the final A mark can be awarded. Regard a power of ten error as an arithmetic error.
Transcription errors	If the only error in arriving at a final answer is because given or previously calculated data has clearly been misread but used correctly, all but the final A mark can be awarded.
Fractions	Only accept these where specified in the mark scheme.
Crossed-out work	Work which has been crossed out and not replaced but can easily be read, should be marked as if it had not been crossed out.

Paper 2 example questions mark scheme

Question	Answer	Marks
1(a)	Mercury	B1
1(b)	a moon orbits a planet	B1
	a planet orbits a star / the Sun	B1
1(c)	any two from: comet comes from far away / outside the solar system / highly elliptical orbit / long period / contains ice / has a tail (as it passes the Sun) / loses mass as it passes close to the Sun / comets come in smaller range of sizes / many more asteroids than comets / asteroid made of rock / asteroid made of metals / mostly found between Mars and Jupiter / formed close to the Sun / short period / more circular orbit Candidates must give a valid comparison between a comet and an asteroid in each case, e.g. a comet has a tail but an asteroid does not.	B2
1(d)	(Earth) moves slower	B1
	circumference of orbit / distance travelled in orbit is larger	B1
1(e)(i)	(approximately) 500 s Accept 8 minutes	B1
1(e)(ii)	$(d =) \text{ speed} \times \text{time}$ or $3.0 \times 10^8 \text{ (m / s)}$ seen	B1
	$1.5 \times 10^{11} \text{ (m)}$	A1

Question	Answer	Marks
2(a)(i)	the Earth to travel round the Sun	B1
2(a)(ii)	the Moon to travel round the Earth	B1
2(a)(iii)	the Earth to rotate <u>once</u> (on its own axis) / the Earth to complete one rotation	B1
2(b)	$(d =) vT$ in any form algebraic or numerical, e.g. $(r = \frac{vT}{2\pi})$	C1
	$3.5 \times 10^4 \times 29 \times 365 \times 24$	C1
	$1.4 \times 10^9 \text{ (km)}$	A1

PRACTICE

from 2023

Question	Answer	Marks
2(c)(i)	Saturn has rotated once and Earth 29 times	B1
	Jupiter has rotated 2.4 times (in 29 years) / Jupiter is now less than halfway (round its orbit)	B1
2(c)(ii)	Earth on line to right	B1
	Jupiter on line to left	B1
	Saturn about 75° clockwise, not quite on line downwards	B1

Question	Answer	Marks
3(a)	Venus	B1
	Mars	B1
3(b)	smallest distance 6.3×10^8 km	B1
	largest distance 9.3×10^8 km	B1
3(c)(i)	reduces at t_1 and has a flat section somewhere between t_1 and t_2	B1
	after t_2 back to initial level and neither change at t_1 or t_2 is vertical	B1
3(c)(ii)	lower brightness when light is blocked / flat section at a lower level	B1
	time to reduce brightness and time to increase brightness is longer / flat section for a shorter time	B1
3(c)(iii)	from t_1 to the next time when the brightness starts to reduce or from t_2 to the next time when the brightness starts to increase	B1

Question	Answer	Marks
4(a)(i)	infrared and ultraviolet	B1
4(a)(ii)	(end with) the longest wavelength or lowest frequency	B1
4(b)(i)	collection of stars	C1
	billions of stars	A1
4(b)(ii)	use of 3.0×10^8 m / s	B1
	365 (days in a year) and 24 (hours in a day) seen	B1
	$3.0 \times 10^8 \times 365 \times 24 \times 60 \times 60$ or 9.46×10^{15} seen	B1

Question	Answer	Marks
4(b)(iii)	further from Earth the greater the speed of recession or speed of recession proportional to distance from Earth	B1
4(b)(iv)	values are (nearly) the same	B1
	all galaxies (and Earth) were together (13–14 billion years ago)	B1

Question	Answer	Marks
5(a)(i)	(hydrogen) nuclei come together / fuse	B1
	to form helium and release energy	B1
5(a)(ii)	high speeds needed	B1
	nuclei repel	B1
5(b)(i)	becomes a main sequence star / stable star	B1
	expands to become a red supergiant	B1
	becomes a supernova (and throws off outer layers)	B1
	core becomes a neutron star or (if original mass large enough) core shrinks to become a black hole	B1
5(b)(ii)	in a supernova	B1

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